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Science-fiction

DECEMBER 1944

25 CENTS

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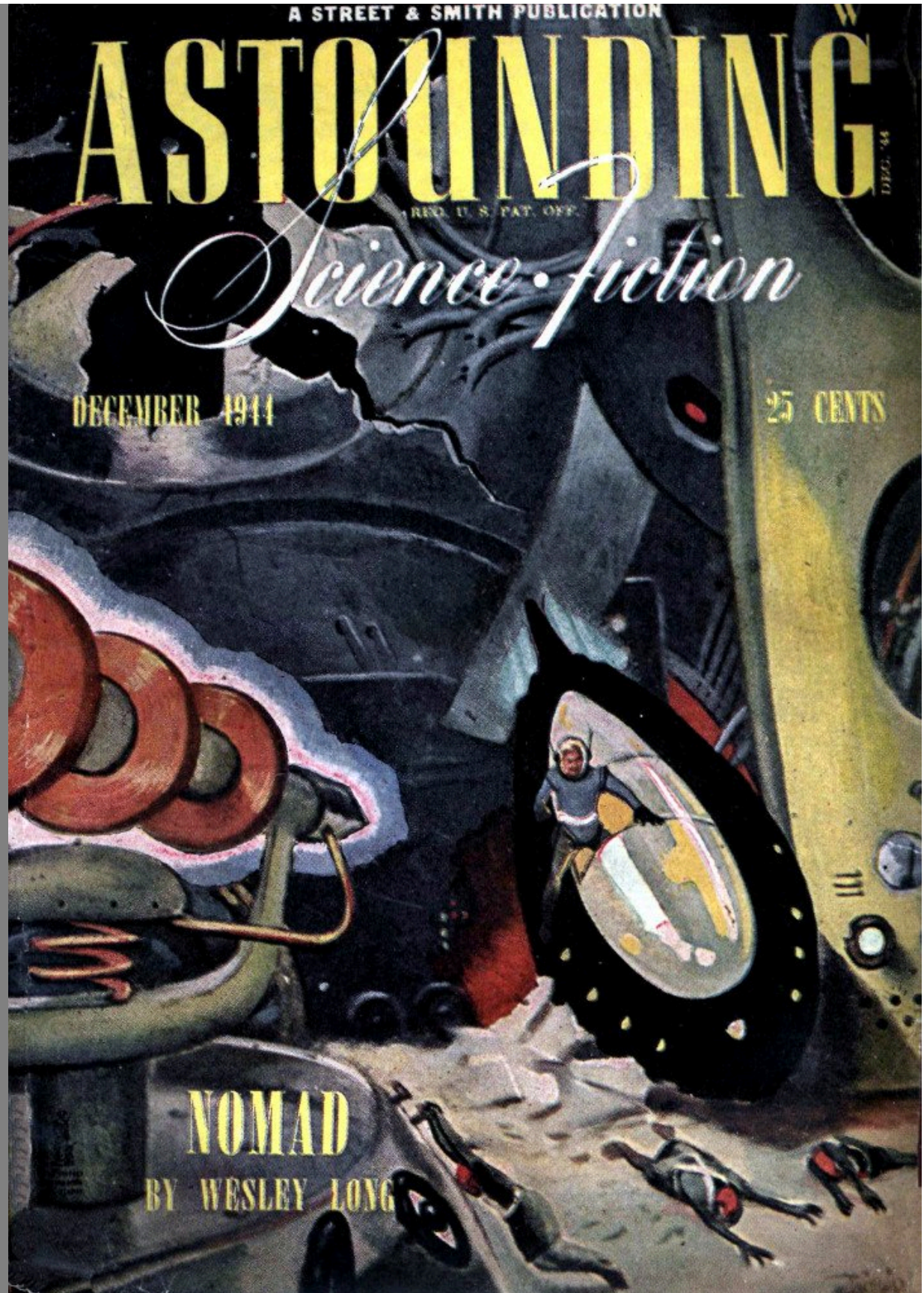
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*** START OF THE PROJECT GUTENBERG EBOOK THE FIRING
LINE ***

Firing Line

By GEORGE O. SMITH

Illustrated by Orban

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Mark Kingman was surprised by the tapping on his windowpane. He thought that the window was unreachable from the outside—and then he realized that it was probably someone throwing bits of dirt or small stones. But who would do that when the doorway was free for any bell-ringer?

He shrugged, and went to the window to look out—and become cross-eyed as his eyes tried to cope with a single circle not more than ten inches distant. He could see the circle—and the bands on the inside spiraling into the depths of the barrel, and a cold shiver ran up his spine from there to here. Behind the heavy automatic, a dark-complected man with a hawklike face grinned mirthlessly.

Kingman stepped back and the stranger swung in and sat upon the windowsill.

"Well?" asked the lawyer.

"Is it well?" asked the stranger. "You know me?"

"No. Never saw you before in my life? Is this a burglary?"

"Nope. If it were, I'd have drilled you first so you couldn't describe me."

Kingman shuddered. The stranger looked as though he meant it.

"In case you require an introduction," said the hard-faced man, "I'm Allison Murdoch."

"Hellion?"

"None other."

"You were in jail—"

"I know. I've been there before."

"But how did you escape?"

"I'm a doctor of some repute," said Hellion, "Or was, until my darker reputation exceeded my reputation for neural surgery. It was simple. I slit my arm and deposited therein the contents of a cigarette. It swelled up like gangrene and they removed me to the hospital. I removed a few guards and lit out in the ambulance. And I am here."

"Why?" Kingman then became thoughtful. "You're not telling me this for mutual friendship, Murdoch. What's on your mind?"

"You were in the clink, too. How did you get out?"

"The court proceedings were under question for procedure. It was further ruled that—"

"I see. You bought your way out."

"I did not—"

"Kingman, you're a lawyer. A smart one, too."

"Thank you—"

"But you're capable of buying your freedom, which you did. Fundamentally, it makes no difference whether you bribe a guard to look the other way or bribe a jury to vote the other way. It's bribery in either case."

Kingman smiled in a superior way. "With the very important difference that the latter means results in absolute freedom. Bribing a guard is freedom only so long as the law may be avoided."

"So you did bribe the jury?"

"I did nothing of the sort. It was a ruling over a technicality that did me the favor."

"You created the technicality."

"Look," said Kingman sharply. "You didn't come here to steal by your own admission and your excellent logic. You never saw me before, and I do not know of you save what I've heard. Revenge for something real or fancied is obviously no reason for this visit. I was charged with several kinds of larceny, which charges fell through and I was acquitted of them—which means that I did not commit them. I, therefore, am no criminal. On the other hand, you have a record. You were in jail, convicted, and you escaped by some means that may have included the act of first-degree murder. You came here for some reason, Murdoch. But let me tell you this: I am in no way required to explain the workings of my mind. If you expect me to reveal some legal machination by which I gained my freedom, you are mistaken. As far as the solar system is concerned, everything was legal and above board."

"I get it," smiled Murdoch. "You're untouchable."

"Precisely. And rightfully so."

"You're the man I want, then."

"It isn't mutual. I have no desire to be identified with a criminal of your caliber."

"What's wrong with it?" asked Murdoch.

"It is fundamentally futile. You are not a brilliant criminal. You've been caught."

"I didn't have the proper assistance. I shall not be caught again. Look," he said suddenly, "how is your relationship with Venus Equilateral?"

Kingman gritted his teeth and made an animal noise.

"I thought so. I have a score of my own to settle. But I need your help. Do I get it?"

"I can't see how one of your caliber is capable—"

"Are you or aren't you? Your answer may decide the duration of your life."

"You needn't threaten. I'm willing to go to any lengths to get even with Channing and his crowd. But it must be good."

"I was beaten by a technical error," explained Murdoch. "The coating on my ship did it."

"How?"

"They fired at me with a super electron-gun. A betatron. It hit me and disrupted the ship's apparatus. The thing couldn't have happened if the standard space-finish hadn't been applied to the *Hippocrates*."

"I'm not a technical man," said Kingman. "Explain, please."

"The average ship is coated with a complex metallic oxide which among other things inhibits secondary emission. Had we been running a ship without this coating, the secondary emission would have left the

Hippocrates in fair condition electronically, but the Relay Station would have received several times the electronic charge. But the coating accepted the terrific charge and prevented the normal urge of electrons to leave by secondary emission—"

"What is secondary emission?"

"When an electron hits at any velocity, it drives from one to as high as fifty electrons from the substance it hits. The quantity depends upon the velocity of the original electron, the charges on cathode and anode, the material from which the target is made, and so on. We soaked 'em in like a sponge and took it bad. But the next time, we'll coat the ship with the opposite stuff. We'll take a bit of Venus Equilateral for ourselves."

"I like the idea. But how?"

"We'll try no frontal attack. Storming a citadel like Venus Equilateral is no child's play, Kingman. As you know, they're prepared for anything either legal or technical. I have a great respect for the combined abilities of Channing and Franks. I made my first mistake by giving them three days to make up their minds. In that time, they devised, tested, and approved an electron weapon of some power. Their use of it was as dangerous to them as it was to me—or would have been if I'd been prepared with a metallic-oxide coating of the proper type."

"Just what are you proposing?" asked Kingman. "I do not understand what you're getting at."

"You are still one of the officials of Terran Electric?"

"Naturally."

"You will be surprised to know that I hold considerable stock in that company."

"How, may I ask?"

"The last time you bucked them, you did it on the market. You lost," grinned Murdoch. "Proving that you haven't a one hundred percent record either. Well, while Terran Electric was dragging its par value down around the twos and threes, I took a few shares."

"How do you stand?"

"I rather imagine that I hold fifteen or twenty percent."

"That took money."

"I have money," said Murdoch modestly. "Plenty of it. I should have grabbed more stock, but I figured that between us we have enough to do as we please. What's your holdings?"

"I once held forty-one percent. They bilked me out of some of that. I have less than thirty percent."

"So we'll run the market crazy again, and between us we'll take off control. Then, Kingman, we'll use Terran Electric to ruin Venus Equilateral."

"Terran Electric isn't too good a company now," admitted Kingman. "The public stays away in huge droves since we bucked Interplanetary Communications. That bunch of electronic screwballs has the public acclaim. They're now in solid since they opened person-to-person phone on the driver frequencies. You can talk to someone in the Palanortis Country of Venus with the same quality and speak-ability that you get in making a call from here to the house across the street."

"Terran Electric is about finished," said Murdoch flatly. "They shot their wad and lost. You'll be bankrupt in a year, and you know it."

"That includes you, doesn't it?"

"Terran Electric is not the mainstay of my holdings," smiled Murdoch. "Under assumed names, I have picked up quite a few bits. Look, Kingman, I'm advocating piracy!"

"Piracy?" asked Kingman aghast.

"Illegal piracy. But I'm intelligent. I realize that a pirate hasn't a chance against civilization unless he is as smart as they are. We need a research and construction organization, and that's where Terran Electric comes in. It's an old company, well established. It's now on the rocks. We can build it up again. We'll use it for a base, and set the research boys to figuring out the answers we need. Eventually we'll control Venus Equilateral, and half of the enterprises throughout the system."

"And your main plan?"

"You run Terran Electric, and I'll run the space piracy. Between us we'll have the system over a barrel. Space craft are still run without weapons, and no weapons are suited for space fighting. But the new field opened up by the driver radiation energy may exhibit something new in weapons. That's what I want Terran Electric to work on."

"We'll have to plan a bit more," said Kingman thoughtfully. "I'll cover you up, and eventually we'll buy you out. Meanwhile we'll go to work on the market and get control of Terran Electric. And plan, too. It'll have to be foolproof."

"It will be," said Murdoch. "We'll plan it that way."

"We'll drink on it," said Kingman.

"*You'll* drink on it," said Murdoch. "I never touch the stuff. I still pride myself on my skill with a scalpel, and I do not care to lose it. Frankly, I hope to keep it long enough to uncover the metatarsal bones of one Donald Channing, Director of Communications."

Kingman shuddered. At times, murder had passed through his mind when thinking of Channing. But this cruel idea of vivisecting an enemy indicated a sadism that was far beyond Kingman's idea of revenge. Of course, Kingman never considered that ruining a man financially, reducing him to absolute dependency upon friends or government, when the man had spent his life in freedom and plenty—the latter gained by his ability under freedom—was cruel and inhuman.

And yet it would take a completely dispassionate observer to tell which was worse; to ruin a man's body or to ruin a man's life.

The man in question was oblivious to these plans on his future. He was standing before a complicated maze of laboratory glassware and a haywire tangle of electronic origin. He looked it over in puzzlement, and his lack of enthusiasm bothered the other man. Wesley Farrell thought that his boss would have been volubly glad to see the fruits of his labor.



"No doubt it's wonderful," smiled Channing. "But what is it, Wes?"

"Why, I've been working on an alloy that will not sustain an arc."

"Go on. I'm interested even though I do not climb the chandelier and scream, beating my manly chest."

"Oil switches are cumbersome. Any other means of breaking contact is equally cumbersome if it is to handle much power. My alloy is non-arcing. It will not sustain an arc, even though the highest current and voltage are broken."

"Now I am really interested," admitted Channing. "Oil switches in a spaceship are a definite drawback."

"I know. So—here we are."

"What's the rest of this stuff?" asked Channing, laying a hand on the glassware.

"Be careful!" said Farrell in concern. "That's hot stuff."

"Oh?"

"In order to get some real voltages and currents to break without running the main Station bus through here, I cooked this stuff up. The plate-grilleworks in the large tubes exhibit a capacity between them of about one microfarad. Empty, that is, or I should say precisely point nine eight microfarads in vacuo. The fluid is of my own devising, concocted for the occasion, and has a dielectric constant of thirteen times ten to the sixth power. It—"

"Great Howling Rockets!" exploded Channing. "That makes the overall capacity equal to thirteen *farads*!"

"Just about. Well, I have the condenser charged to three kilovolts, and then I discharge it through this switch made of the non-arcing alloy. Watch! No, Don, from back here, please, behind this safety glass."

Channing made some discomfoting calculations about thirteen farads at three thousand volts charge and decided that there was something definitely unlucky about the number thirteen.

"The switch, now," continued Farrell, as though thirteen farads was just a mere drop in the bucket, "is opened four milliseconds after it is closed. The time-constant of the discharging resistance is such that the voltage is point eight three, of its peak three thousand volts, giving a good check of the alloy."

"I should think so," groused Channing. "Eighty-three percent of three thousand volts is just shy of twenty-five hundred volts. The current of discharge passing through a circuit that will drop the charge in a thirteen farad condenser eighty-three percent in four milliseconds will be something fierce, believe me."

"That is why I use the heavy busbars from the condenser bank through the switch."

"I get it. Go ahead, Wes. I want to see this non-arcing switch of yours perform."

Farrell checked the meters, and then said "Now!" and punched the switch at his side. Across the room a solenoid drove the special alloy bar between two clamps of similar metal. Almost immediately, four thousandths of a second later, to be exact, the solenoid reacted automatically and the no-arc alloy was withdrawn. A minute spark flashed briefly between the contacts.

"And that is that," said Channing, slightly dazed by the magnitude of it all, and the utter simplicity of the effects. "But look, Wes, may I ask you a favor? Please discharge that infernal machine and drain that electrolyte out. Then make the thing up in a tool-steel case and seal it. Also hang on busbars right at the plates themselves, and slap a peak-voltage fuse across the terminals. One that will close at anything above three thousand volts. Follow me?"

"I think so. But that is not the main point of interest—"

"I know," grinned Channing, mopping his forehead. "The non-arc is. But that fragile glassware makes me as jittery as a Mexican jumping bean."

"But why?"

"Wes, if that glassware fractures somewhere, and that electrolyte drools out, you'll have a condenser of one microfarad—charged to thirteen million times three thousand volts. Or, in nice, hollow, round numbers, forty billion volts! Four times ten to the tenth. Of course, it won't get that far. It'll arc across the contacts before it gets that high, but it might raise particular hell on the way out. Take it easy, Wes. We're seventy millionodd miles from the nearest large body of dirt, all collected in a little steel bottle about three miles long and a mile in diameter. I'd hate to stop all interplanetary communications while we scraped ourselves off of the various walls and treated ourselves for electric shock. It would—the discharge itself, I mean—raise hell with the equipment anyway. So play it easy, Wes. We do not permit certain experiments out here because of the slow neutrons that sort of wander through here at fair density. Likewise, we cannot permit dangerous experiments. And anything that includes a dangerous experiment must be out, too."

"Oh," said Wes. His voice and attitude were together crestfallen.

"Don't take it so hard, fella," grinned Channing. "Anytime we have to indulge in dangerous experiments, we always do it with an assistant—and in

one of the blister-laboratories. But take that fragile glassware out of the picture and I'll buy it," he finished.

Walt Franks entered and asked what was going on.

"Wes was just demonstrating the latest equipment in concentrated deviltry," smiled Channing.

"That's my department," said Walt.

"Oh, it's not as bad as your stuff," said Channing. "What he's got here is an alloy that will break several million watts without an arc. Great stuff, Walt."

"Sounds swell," said Walt. "Better scribble it up and we'll get a patent. It sounds useful."

"I think it may bring us a bit of change," said Channing. "It's great stuff, Wes."

"Thanks. It annoyed me to see those terrific oil-breakers we have here. All I wanted to do was to replace 'em with something smaller and more efficient."

"You did, Wes. And that isn't all. How did you dream up that high-dielectric?"

"Applied several of the physical phenomena."

"That's a good bet, too. We can use several fluids of various dielectric constants. Can you make solids as well?"

"Not as easily. But I can try—?"

"Go ahead and note anything you find above the present, listed compounds and their values."

"I'll list everything, as I always do."

"Good. And the first thing to do is to can that stuff in a steel case."

"It'll have to be plastalloy."

"That's as strong as steel and nonconducting. Go ahead."

Channing led Franks from the laboratory, and once outside Channing gave way to a session of the shakes. "Walt," he asked plaintively, "take me by the hand and lead me to Joe's. I need some vitamins."

"Bad?"

"Did you see that glassblower's nightmare?"

"You mean that collection of cut glass?" grinned Walt. "Uh-huh. It looked as though it were about to collapse of its own dead weight."

"That held an electrolyte of dielectric constant thirteen times ten to the sixth. He had it charged to a mere three thousand volts. Ye Gods, Walt. Thirteen farads at three KV. *Whew*. And when he discharged it, the confounded leads that went through the glass sidewalls to the condenser plates positively glowed in the cherry red. I swear it!"

"He's like that," said Walt. "You shouldn't worry about him. He'll have built that condenser out of good stuff—the leads will be alloys like those we use in the bigger tubes. They wouldn't fracture the glass seals no matter what the temperature difference between them and the glass was. Having that alloy around the place—up in the tube-maintenance department they have a half ton of quarter-inch rod—he'd use it naturally."

"Could be, Walt. Maybe I'm a worry wart."

"You're not used to working with his kind."

"I quote: 'Requiring a high voltage source of considerable current capacity, I hit upon the scheme of making a super-high capacity condenser and discharging it through my no-arc alloy. To do this it was necessary that I invent a dielectric material of C equals thirteen times ten to the sixth.' Unquote."

"Wes is a pure scientist," reminded Walt. "If he were investigating the electrical properties of zinc, and required solar power magnitudes to complete his investigation, he'd invent it and then include it as incidental to the investigation on zinc. He's never really understood our recent divergence in purpose over the power tube. That we should make it soak up power from Sol was incidental and useful only as a lever or means to make Terran Electric give us our way. He'd have forgotten it, I'll bet, since it was not the ultimate goal of the investigation."

"He knows his stuff, though."

"Granted. Wes is brilliant. He is a physicist, though, and neither engineer nor inventor. I doubt that he is really interested in the practical aspects of anything that is not directly concerned with his eating and sleeping."

"What are we going to do about him?"

"Absolutely nothing. You aren't like him—"

"I hope not."

"And conversely, why should we try to make him like you?"

"That I'm against!" chimed in a new voice. Arden Channing took each man by the arm and looked up on either side of her, into one face and then the other. "No matter how, why, when, who, or what, one like him is all that the solar system can stand."

"Walt and I are pretty much alike."

"Uh-huh. You are. That's as it should be. You balance one another nicely. You couldn't use another like you. You're speaking of Wes Farrell?"

"Right."

"Leave him alone," said Arden sagely. "He's good as he is. To make him similar to you would be to spoil a good man. He'd then be neither fish, flesh, nor fowl. He doesn't think as you do, but instead proceeds in a straight line from remote possibility to foregone conclusion. Anything that gets needed en route is used, or gadgeteered and forgotten. That's where you come in, fellows. Inspect his by-products. They may be darned useful."

"O.K. Anybody care for a drink?"

"Yup. All of us," said Arden.

"Don, how did you rate such a good-looking wife?"

"I hired her," grinned Channing. "She used to make all my stenographic mistakes, remember?"

"And gave up numerous small errors for one large one? Uh-huh. I recall. Some luck."

"It was my charm."

"Baloney. Arden, tell the truth. Didn't he threaten you with something terrible if you didn't marry him?"

"You tell him," grinned Channing. "I've got work to do."

Channing left the establishment known as Joe's and advertised as the "Best bar in twenty-seven million miles, minimum," and made his way toward his office slowly. He didn't reach it. Not right away. He was intercepted by Charley Thomas who invited him to view a small experiment. Channing smiled and said that he'd prefer to see an experiment of any kind to going to his office, and followed Charley.

"You recall the gadget we use to get perfect tuning with the alloy-selectivity transmitter?"

"You mean that variable alloy disk all bottled up and rotated with a selsyn?" asked Don, wondering what came next. "Naturally I remember it. Why?"

"Well, we've found that certain submicroscopic effects occur with inert objects. What I mean is this: Given a chunk of cold steel of goodly mass and tune your alloy disk to pure steel, and you can get a few micro-microamperes output if the tube is pointed at the object."

"Sounds interesting. How much amplification do you need to get this reading and how do you make it tick?"

"We run the amplifier up to the limit and then sweep the tube across the object sought, and the output meter leaps skyward by just enough to make us certain of our results. Watch!"

Charley set the tube in operation and checked it briefly. Then he took Don's hand and put it on the handle that swung the tube on its gimbals. "Sort of paint the wall with it," he said. "You'll see the deflection as you pass the slab of tool steel that's standing there."

Channing did, and watched the minute flicker of the ultra-sensitive meter. "Wonderful," he grinned, as the door opened and Franks entered.

"Hi, Don. Is it true that you bombarded her with flowers?"

"Nope. She's just building up some other woman's chances. Have you seen this effect?"

"Yeah—it's wonderful, isn't it?"

"That's what I like about this place," said Charley with a huge smile. "That's approximately seven micro-microamperes output after amplification on the order of two hundred million times. We're either working on something so small we can't see it or something so big we can't count it. It's either fifteen decimal places to the left or to the right. Every night when I go home, I say a little prayer. I say: 'Dear God, please let me find something today that is based upon unity, or at least no more than two decimal places' but it is no good. If He hears me at all, He's too busy to bother with things that the human race classifies as 'One.'"

"How do you classify resistance, current, and voltage?" asked Channing, manipulating the tube on its gimbals and watching the effect.

"One million volts across ten megohms equals one hundred thousand microamperes. That's according to Ohm's Law."

"He's got the zero-madness too," chuckled Walt. "It obtains from thinking in astronomical distances, with interplanetary coverages in watts, and celestial input, and stuff like that. Don, this thing may be handy, some day. I'd like to develop it."

"I suggest that couple of stages of tube-amplification might help. Amplify it before transduction into electronic propagation."

"We can get four or five stages of sub-electronic amplification, I think. It'll take some working."

"O.K., Charley. Cook ahead. We do not know whither we are heading, but it looks darned interesting."

"Yeah," added Walt, "it's a darned rare scientific fact that can't be used for something, somewhere. Well, Don, now what?"

"I guess we now progress to the office and run through a few reams of paper-work. Then we may relax."

"O.K. Sounds good to me. Let's go."

Hellion Murdoch pointed to the luminous speck in the celestial globe. His finger stabbed at the marker button, and a series of faint concentric spheres marked the distance from the center of the globe to the object, which Murdoch read and mentioned: "Twelve thousand miles."

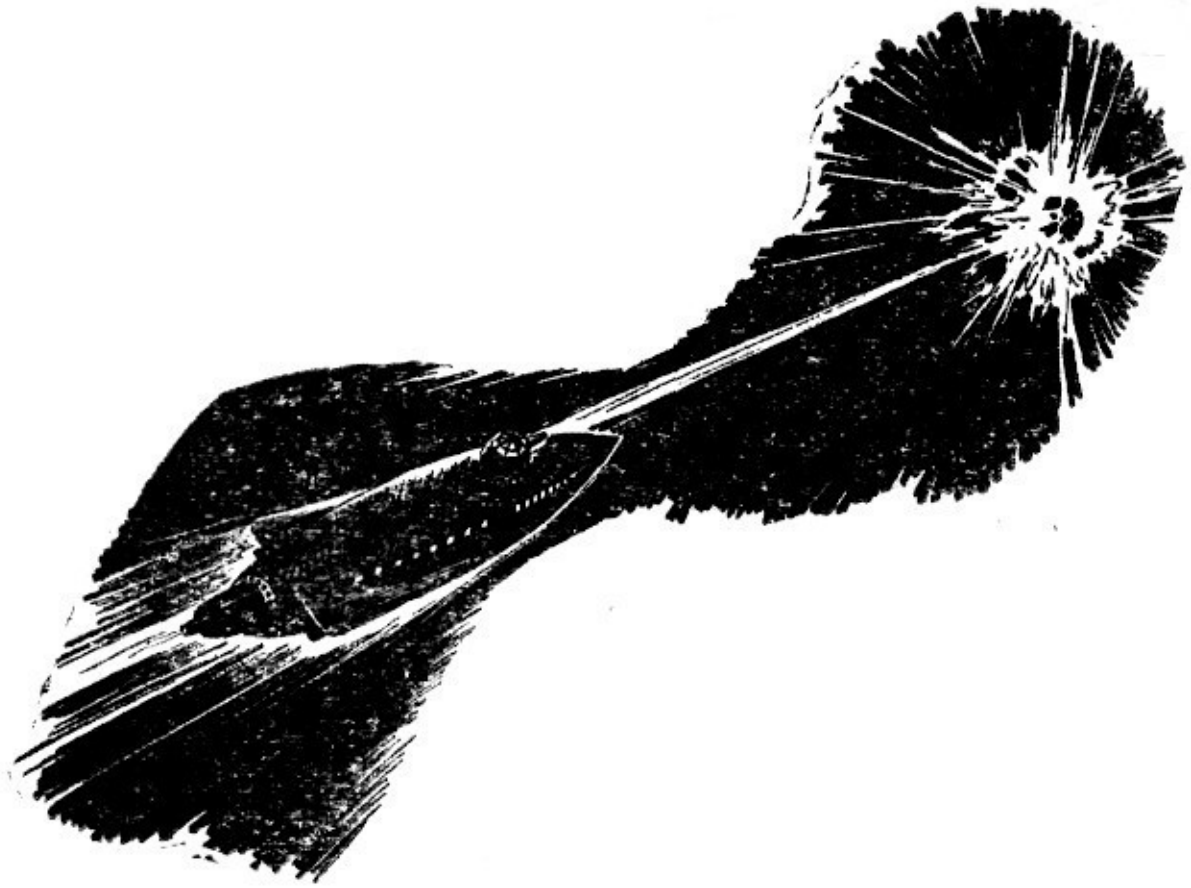
"Asteroid?" asked Kingman.

"What else?" asked Murdoch. "We're lying next to the Asteroid Belt."

"What are you going to do?"

"Burn it," said Murdoch. His fingers danced upon the keyboard, and high above him, in the dome of the *Black Widow*, a power intake tube swiveled and pointed at Sol. Coupled to the output of the power intake tube, a power-output tube turned to point at the asteroid. And Murdoch's poised finger came down on the last switch, closing the final circuit.

Meters leaped up across their scales as the intangible beam of solar energy came silently in and went as silently out. It passed across the intervening miles with the velocity of light squared, and hit the asteroid. A second later the asteroid glowed and melted under the terrific bombardment of solar energy directed in a tight beam.



"It's O.K.," said Hellion. "But have the gang build us three larger tubes to be mounted turretwise. Then we can cope with society."

"What do you hope to gain by that? Surely piracy and grand larceny are not profitable in the light of what we have and know."

"I intend to institute a reign of terror."

"You mean to go through with your plan?"

"I am a man of my word. I shall levy a tax against each and every ship leaving any spaceport. We shall demand one dollar solarian for every gross ton that lifts from any planet and reaches the planetary limit."

"How do you establish that limit?" asked Kingman interestedly.

"Ironically, we'll use the Channing Layer," said Murdoch with dark humor. "Since the Channing Layer describes the boundary below which our solar beam will not work. Our reign of terror will be identified with Channing because of that; it will take some of the praise out of people's minds when they think of Channing and Interplanetary Communications."

"That's pretty deep psychology," said Kingman.

"You should recognize it," smiled Murdoch. "That's the kind of stuff you legal lights pull. Mention the accused in the same sentence with one of the honored people; mention the defendant in the same breath with one of the hated people—it's the same stunt. Build them up or tear them down by reference."

"You're pretty shrewd."

"I am," agreed Murdoch placidly.

"Mind telling me how you found yourself in the fix you're in?"

"Not at all. I've been interested for years in neuro-surgery. My researches passed beyond the realm of rabbits and monkeys, and I found it necessary to investigate the more delicate, more organized, the higher-strung. That means human beings—though some of them are less sensitive than a rabbit and less delicate than a monkey." Murdoch's eyes took on a cynical expression at this. Then it passed and he continued: "I became famous, as you know. Or do you?"

Kingman shook his head.

"I suppose not. I became famous in my own circle. Lesser neuro-surgeons sent their complex cases to me; unless you were complex, you would never hear of Allison Murdoch. Well anyway, some of them offered exciting opportunities. I—frankly, experimented. Some of them died. It was quite a bit of cut and try because not too much has been written on the finer points of the nervous system. But there were too few people who were complex enough to require my services, and I turned to clinical work, and experimented freely."

"And there you made your mistake?"

"Do you know how?"

"No. I imagine that with many patients you exceeded your rights once too often."

"Wrong. It is a funny factor in human relationship. Something that makes no sense. When people were paying me three thousand dollars an hour for operations, I could experiment without fear. Some died, some regained their health under my ministrations. But when I experimented on charity patients, I could not experiment because of the 'Protection' given the poor. The masses were not to be guinea pigs. Ha!" laughed Murdoch, "only the rich are permitted to be subjects of an experiment. Touch not the poor, who offer nothing. Experiment upon those of intellect, wealth, fame, or anything that sets them above the mob. Yes, even genius came under my knife. But I couldn't give a poor man a fifty-fifty chance at his life, when the chances of his life were less than one in ten. From a brilliant man, operating under fifty-fifty chances for life, I became an inhuman monster that cut without fear. I was imprisoned, and later escaped with some friends."

"And that's when you stole the *Hippocrates* and decided that the solar system should pay you revenge-money?"

"I would have done better if I had not made that one mistake. I forgot that in the years of imprisonment, I fell behind in scientific knowledge. I know now that no one can establish anything at all without technical minds behind him."

Kingman's lips curled. "I wouldn't agree to that."

"You should. Your last defeat at the hands of the technicians you scorn should have taught you a lesson. If you had been sharp, you would have outguessed them; outengineered them. They, Kingman, were not afraid to rip into their detector to see what made it tick."

"But I had only the one—"

"They knew one simple thing about the universe. That rule is that if anything works once, it may be made to work again." He held up his hand as Kingman started to speak. "You'll bring all sorts of cases to hand and try to disprove me. You can't. Oh, you couldn't cause a quick return of the diplodocus, or re-enact the founding of the solar government, or even reburn

a ton of coal. But there is other carbon, there will be other governmental introductions and reforms, and there may some day be the rebirth of the dinosaur—on some planet there may be carboniferous ages now. Any phenomena that is a true phenomena—and your detector was definite, not a misinterpretation of effect—can be repeated. But, Kingman, we'll not be outengineered again."

"That I do believe."

"And so we will have our revenge on Interplanetary Communications and upon the system itself."

"We're heading home now?"

"Right. We want this ship fitted with the triple turret I mentioned before. Also I want the interconnecting links between the solar intake and the power-projectors beefed up. When you're passing several hundred megawatts through any system, losses of the nature of .000,000,1% cause heating to a dangerous degree. We've got to cut the I^2R losses. I gave orders that the turret be started, by the way. It'll be almost ready when we return."

"You gave orders?" said Kingman.

"Oh yes," said Hellion Murdoch with a laugh. "Remember our last bout with the stock market? I seem to have accumulated about forty-seven percent. That's sufficient to give me control of our company."

"But ... but—" spluttered Kingman. "That took money—"

"I still have enough left," said Murdoch quietly. "After all, I spent years in the Melanortis Country of Venus. I was working on the *Hippocrates* when I wasn't doing a bit of mining. There's a large vein of platiniridium there. You may answer the rest."

"I still do not get this piracy."

Murdoch's eyes blazed. "That's my interest. That's my revenge! I intend to ruin Don Channing and Venus Equilateral. With the super turret they'll never be able to catch us, and we'll run the entire system."

Kingman considered. As a lawyer, he was finished. His last try at the ruination of the Venus Equilateral crowd by means of pirating the interplanetary communications beam, well that was strictly a violation of the Communications Code. The latter absolutely prevented any man or group of men from diverting communications not intended for them and using these communications for their own purpose. His defense that Venus Equilateral had also violated the law went unheard. It was pointed out to him that Venus Equilateral tapped his own line, and the tapping of an illegal line was the act of a communications agent in the interest of the government. He was no longer a lawyer, and in fact he had escaped a long jail term by sheer bribery.

He was barred from legal practice, and he was barred from any business transactions. The stock market could be manipulated, but only through a blind, which was neither profitable nor safe.

His holdings in Terran Electric was all that stood between him and ruin. He was no better off than Murdoch, save that he was not wanted.

But—

"I'm going to remain on Terra and run Terran Electric like a model company," he said. "That'll be our base."

"Right. Except for a bit of research along specified lines, you will do nothing. Your job will be to act apologetic for your misdeeds. You will grovel on the floor before any authority, and beseech the legal profession to accept you once more. I will need your help, there. You are to establish yourself in the good graces of the Interplanetary Patent Office and report to me any applications that may be of interest. The research that Terran Electric will conduct will be along innocuous lines. The real research will be conducted in a secret laboratory. The one in the Melanortis Country. Selected men will work there, and the Terran Electric fleet of cargo-carriers will carry the material needed. My main failure was not to have provided a means of knowing what the worlds were doing. I'll have that now, and I shall not be defeated again."

"We'll say that one together!" said Kingman. He flipped open a large book and set the autopilot from a set of figures. The *Black Widow* turned gently and started to run for Terra at 2-G.

Walt Franks frowned at the memorandum in his hand. "Look, Don, are we ever going to get to work on that deal with Keg Johnson?"

"Uh-huh," answered Don, without looking up.

"He's serious. Transplanet is getting the edge, and he doesn't like it."

"Frankly, I don't like dabbling in stuff like that either. But Keg's an old friend, and I suppose that's how a guy gets all glommed up on projects, big business deals, and so forth. We'll be going in directly. Why the rush?"

"A bit of personal business on Mars which can best be done at the same time, thus saving an additional trip."

"O.K.," said Don idly. "Might as well get it over with. Can you pack in an hour?"

"Sure. I'll be there."

Actually, it was less than an hour before the *Relay Girl* went out of the South End Landing Stage, turned, and headed for Mars. Packing to the Channings was a matter of persuading Arden not to take everything but the drapes in the apartment along with her, while for Walt Franks it was a matter of grabbing a trunkful of instruments and spare parts. Space travel is a matter of waiting for days in the confines of a small bubble of steel. Just waiting. For the scenery is unchanging all the way from Sol to Pluto—and is the same scenery that can be seen from the viewports of Venus Equilateral. Walt enjoyed his waiting time by tinkering; having nothing to do would have bored him, and so he took with him enough to keep him busy during the trip.

At two Terran gravities, the velocity of the *Relay Girl* built up bit by bit and mile by mile until they were going just shy of one thousand miles per second. This occurred an hour before turnover, which would take place at the twenty-third hour of flight.

And at that time there occurred a rarity. Not an impossibility like the chances of collision with a meteor, those things happen only once in a lifetime, and Channing had had his collision. Nor was it as remote as getting a royal flush on the deal. It happened, not often, but it did happen to some ships occasionally.

Another ship passed within detector range.

The celestial globe glimmered faintly and showed a minute point at extreme range. Automatic marker-spheres appeared concentrically within the celestial globe and colures and diameters marked the globe off into octants. A dim red line appeared before the object, giving the probable course of the object.

Bells rang briefly, and the automatic meteor-circuits interpreted the orbit of the oncoming object and decided that the object was not dangerous. Then they relaxed. Their work was done until another object came within range for them to inspect. They were no longer interested, and they forgot about the object with the same powers of complete oblivion that they would have exerted on a meteor of nickel and iron.

They were mechanically incapable of original thought. So the object, to them, was harmless.

Channing looked up at the luminescent spot, sought the calibration spheres, made a casual observation, and forgot about it. To him it was a harmless meteor.

Even the fact that his own velocity was a thousand miles per second, and the object's velocity was the same, coming to them on a one hundred and seventy degree course and due to pass within five thousand miles did not register. Their total velocity of two thousand miles did not register just because of that rarity with which ships pass within detector range, while meteors are encountered often.

Had Channing been thinking about the subject in earnest, he would have known—for it is only man, with all too little time, who uses such velocities. The universe, with eternity in which to work her miracle, seldom moves in velocities greater than forty or fifty miles per second.

Channing forgot it, and as the marker-spheres switched to accommodate the approaching object, he turned to more important things.

In the other ship, Hellion Murdoch frowned. He brightened, then, and depressed the plunger that energized his solar beam and projector. He did not recognize the oncoming object for anything but a meteor, either, and his desire was to find out how his invention worked at top speeds.

Kingman asked: "Another one?"

"Uh-huh," said Murdoch idly. "I want to check my finders."

"But they can't miss."

"No? Look, lawyer, you're not running a job that may be given a stay or a reprieve. The finders run on light velocities. The solar beam runs on the speed of light squared. We'll pass that thing at five thousand miles and at two thousand accumulative miles per second. A microsecond of misalignment, and we're missing, see? I think we're going to be forced to put correction circuits in so that the vector sums and velocities and distances will all come out with a true hit. It will not be like sighting down a searchlight beam at high velocity."

"I see. You'll need compensation?"

"Plenty, at this velocity and distance. This is the first time I've had a chance to try it out."

The latter fact saved the *Relay Girl*. By a mere matter of feet, and inches; by the difference between the speed of light and the speed of light squared at a distance of five thousand miles, plus a slight miscompensation. The intolerably hot umbra of Murdoch's beam followed below the pilot's greenhouse of the *Relay Girl* all the way past, a matter of several seconds. The spill-over was tangible enough to warm the *Relay Girl* to uncomfortable temperatures.

Then with no real damage done, the contact with ships in space was over, but not without a certain minimum of recognition.

"Hell!" said Kingman. "That was a space craft."

"Who?"

"I don't know. You missed."

"I'd rather have hit," said Murdoch coldly. "I hope I missed by plenty."

"Why?"

"If we scorched their tails any, there'll be embarrassing questions asked."

"So—?"

"So nothing until we're asked. Even then you know nothing."

In the *Relay Girl*, Channing mopped his forehead. "That was hell itself," he said.

Arden laughed uncertainly. "I thought that it would wait until we got there; I didn't expect hell to come after us."

"What—exactly—happened?" asked Walt, coming into the scanning room.

"That—was a spaceship."

"One of this system's?"

"I wonder," said Don honestly. "It makes a guy wonder. It was gone too fast to make certain. It probably was Solarian, but they tried to burn us with something."

"That makes it sound like something alien," admitted Walt. "But that doesn't make good sense."

"It makes good reading," laughed Channing. "Walt, you're the Boy Edison. Have you been tinkering with anything of lethal leanings?"

"You think there may be something powerful afloat?"

"Could be. We don't know everything."

"I've toyed with the idea of coupling a solar intake beam with one of those tubes that Baler and Carroll found. Recall, they smashed up quite a bit of Lincoln Head before they uncovered the secret of how to handle it. Now that we have unlimited power—or are limited only by the losses in our own system—we could, or should be able to, make something rather tough."

"You've toyed with the idea, hey?"

"Uh-huh."

"Of course you haven't really tried it?"

"Of course not."

"How did it work?"

"Fair," grinned Walt. "I did it with miniatures only, of course, since I couldn't get my hooks on a full-grown tube."

"Say," asked Arden, "how did you birds arrive at this idea so suddenly? I got lost at the first premise."

"We passed a strange ship. We heated up to uncomfortable temperatures in a matter of nine seconds flat. They didn't warm us with thought waves, or vector-invectives. Sheer dislike wouldn't do it alone. I guess that someone is trying to do the trick started by our esteemed Mr. Franks here a year or so ago. Only with something practical instead of an electron beam. Honest-to-goodness energy, right from Sol himself, funneled through some tricky inventions. Walt, that experiment of yours. Did you bring it along?"

Walt looked downcast. "No," he said. "It was another one."

"Let's see."

"It's not too good—"

"Same idea?"

Walt went to get his experiment. He returned with a tray full of laboratory glassware, all wired into a maze of electronic equipment.

Channing went white. "You, too?" he yelled.

"Take it easy, sport. This charges only to a hundred volts. We get thirteen hundred microfarads at one hundred volts. Then we drain off the dielectric fluid, and get one billion three hundred million volts charge in a condenser of only one hundred micro-microfarads. It's an idea for the nuclear physics boys. I think it may tend to solidify some of the uncontrollables in the present system of developing high electron velocities."

"That thirteen million dielectric constant stuff is strictly electro-dynamite, I think," said Channing. "Farrell may have developed it as a by-product, but I have a hunch that it will replace some heretofore valuable equipment. The Franks-Farrell generator will outdo Van-Der Graf's little job, I think."

"Franks-Farrell?"

"Sure. He thunk up the dielectric. You thunk up the application. He won't care, and you couldn't have done it without. Follow?"

"Oh sure. I was just trying to figure out a more generic term for it."

"Don't. Let it go as is for now. It's slick, Walt, but there's no weapon in it."

"You're looking for a weapon?"

"Uh-huh. Ever since Murdoch took a swing at Venus Equilateral, I've been sort of wishing that we could concoct something big enough and dangerous enough to keep us free from any other wiseacres. Remember, we stand out there like a sore thumb. We are as vulnerable as a half pound of butter at a banquet for starving Armenians. The next screwball that wants to control the system will have to control Venus Equilateral first. And the best things we can concoct to date include projectile-tossing guns at velocities of less than the speed of our ships, and an electron-shooter that can be overcome by coating the ship with any of the metal-salts that enhance secondary emission."

"Remind me to requisition a set of full-sized tubes when we return. Might as well have some fun."

"O.K., you can have 'em. Which brings us back to the present. Question: Was that an abortive attempt upon our ship, or was that a mistaken try at melting a meteor?"

"I know how to find out. Let's call Charley Thomas and have him get on the rails. We can have him request Terran Electric to give us any information they may have on energy beams to date."

"They'd tell you?" scorned Arden.

"If they write *no!*, and we find out that they did, we'll sue 'em dead. They're too shaky to try anything deep right now."

"Going to make it an official request, hey?"

"Right. From the Station, it'll go out in print, and their answer will be on the 'type, too, since business etiquette requires it. They'll get the implication if they're on the losing end. That'll make 'em try something slick. If they're honest, they'll tell all."

"That'll do it all right," said Walt. "They're too shaky to buck us any more. And if they are trying anything, it'll show."

The rest of the trip was without incident. They put in at Canalopsis and found Keg Johnson with an official 'gram waiting for them. Don Channing

ripped it open and read:

Interplanetary Communications
Attention Dr. Channing:

No project for energy-beam capable of removing meteors under way at Terran Electric, or at any of the subsidiary companies. Ideas suggested along these lines have been disproven by your abortive attempt of a year ago, and will not be considered unless theory is substantiated in every way by practical evidence.

If you are interested, we will delve into the subject from all angles. Please advise.

Terran Electric Co.
Board of Legal Operations
Mark Kingman, LLD.

Channing smiled wryly at Keg Johnson and told him of their trouble.

"Oh?" said Keg, with a frown. "Then you haven't heard?"

"Heard what?"

"Hellion Murdoch has been on the loose for weeks."

"Weeks!" yelled Channing.

"Uh-huh. He feigned gangrene, was taken to the base hospital where he raised hob in his own, inimitable way. He blasted the communications set-up completely, ruined three spaceships, and made off with the fourth. The contact ship just touched there recently and found hell brewing. If they hadn't had a load of supplies and prisoners for the place, they wouldn't have known about it for months, perhaps."

"So! Brother Murdoch is loose again. Well! The story dovetails in nicely."

"You think that was Hellion himself?"

"I'd bet money on it. The official report on Hellion Murdoch said that he was suffering from a very slight persecution complex, and that he was capable of making something of it if he got the chance. He's slightly whacky, and dangerously so."

"He's a brilliant man, isn't he?"

"Quite. His name is well known in the circles of neuro-surgery. He is also known to be an excellent research worker in applied physics."

"Nuts, hey?" asked Walt.

"Yeah, he's nuts. But only in one way, Walt. He's nuts to think that he is smarter than the entire solar system all put together. Well, what do we do now?"

"Butter ourselves well and start scratching for the answer. That betatron trick will not work twice. There must be something."

"O.K., Walt. We'll all help you think. I'm wondering how much research he had to do to develop that beam. After all, we were five thousand miles away, and he heated us up. He must've thought we were a meteor—and another thing, too—he must've thought that his beam was capable of doing something at five thousand miles distance or he wouldn't have tried. Ergo he must have beaten that two hundred mile bugaboo."

"We don't know that the two hundred mile bugaboo is still bugging in space," said Walt, slowly. "That's set up so that the ionization-by-products are not dangerous. Also, he's not transmitting power from station to station, et cetera. He's ramming power into some sort of beam and to the devil with losses external to his equipment. The trouble is, darn it, that we'll have to spend a month just building a large copy of my miniature set-up."

"A month is not too much time," agreed Channing. "And Murdoch will take a swing at us as soon as he gets ready to reach. We can have Charley start building the big tubes immediately, can't we?"

"Just one will be needed. We'll use one of the standard solar intake tubes that we're running the Station from. There's spare equipment aplenty. But the transmitter-terminal tube will take some building."

"Can we buy one from Terran Electric?"

"Why not? Get the highest rating we can. That should be plenty. Terran probably has them in stock, and it'll save us building one."

"What is their highest rating?"

"Two hundred megawatts."

"O.K. I'll send 'em a coded requisition with my answer to their letter."

"What are you going to tell 'em?"

"Tell 'em not to investigate the energy-gun idea unless they want to for their own reasons," Channing grinned. "They'll probably assume—and correctly—that we're going to tinker ourselves."

"And?"

"Will do nothing since it is an extra-planetary proposition. Unless it becomes suitable for digging tunnels, or melting the Martian ice cap," laughed Channing.

Mark Kingman took the letter to Murdoch, who was hidden in the depths of the *Black Widow*. Hellion read it twice, and then growled.

"They smell something, sure," he snarled. "Why didn't we make that a perfect hit!"

"What are we going to do now?"

"Step up our plans. They'll have this thing in a few weeks. Hm-m-m. They order a transmitter-terminal tube. Have you got any in stock?"

"Naturally. Not in stock, but available for the Northern Landing power-line order."

"You have none, then. You will have some available within a few days. That half-promise will stall them from making their own, and every day that they wait for your shipment is a day in our favor. To keep your own nose clean, I'll tell you when to ship the tube. It'll be a few days before I strike."

"Why bother?" asked Kingman. "They won't be around to call names."

"No, but their friends will, and we want to keep them guessing."

"I see. Those tubes are huge enough to excite comment, and there will be squibs in all the papers telling of the giant going to Venus Equilateral, and the Sunday Supplements will all break out in wild guesses as to the reason why Venus Equilateral wants a two hundred megawatt tube. Too bad you couldn't keep your escape a longer secret."

"I suppose so. But it was bound to be out sooner or later anyway. A good general, Kingman, is one whose plans may be changed on a moment's notice without sacrificing. We'll win through."

The days wore on, and the big turret on the top of the *Black Widow* took shape. The supertubes were installed, and Murdoch worked in the bowels of the ship to increase the effectiveness of the course-integrators to accommodate high velocities and to correct for the minute discrepancies that would crop up due to the difference in velocities between light and sub-electronic radiation.



And on Venus Equilateral, the losing end of a war of nerves was taking place. The correspondence by 'type was growing into a reasonable pile, while the telephone conversations between Terran Electric and Venus Equilateral became a daily proposition. The big tubes were not finished. The big tubes were finished, but rejects because of electrode-misalignments. The big tubes were in the rework department. The big tubes were on Luna for their testing. And again they were rejects because the maximum power requirements were not met. They were returned to Evanston and were once more in the rework department. You have no idea how difficult the manufacture of two hundred megawatt tubes really is.

So the days passed, and no tubes were available. The date passed which marked the mythical date of 'if'—*If* Venus Equilateral had started their own manufacturing division on the day they were first ordered from Terran Electric, they would have been finished and available.

Then, one day, word was passed along that the big tubes were shipped. They were on their way, tested and approved, and would be at Venus Equilateral within two days. In the due course of time, they arrived, and the gang at the Relay Station went to work on them.

But Walt Franks shook his head. "Don, we'll be caught like a sitting rabbit."

"I know. But—?" answered Channing.

There was no answer to that question, and so they went to work again.

The news of Murdoch's first blow came that same day. It was a news report from the Interplanetary Network that the Titan Penal Colony had been attacked by a huge black ship of space that carried a huge dome-shaped turret on the top. Beams of invisible energy burned furrows in the frozen ground, and the official buildings melted and exploded from the air pressure within them. The Titan station went off the ether with a roar, and the theorists believed that Murdoch's gang had been augmented by four hundred and nineteen of the Solar System's most vicious criminals.

"That rips it wide open," said Channing. "Better get the folks to prepare to withstand a siege. I don't think they can take us."

"That devil might turn his beams on the Station itself, though," said Walt.

"He wants to control communications."

"With the sub-electron beams we now have, he could do it on far less Station for some time. Not perfectly, but he'd get along."

"Fine future," gritted Channing. "This is a good time to let this project coast, Walt. We've got to start in from the beginning and walk down another track."

"It's easy to say, chum."

"I know it. So far, all we've been able to do is to take energy from the solar intake beams and spray it out into space. It goes like the arrow that went—we know not where."

"So?"

"Forget these gadgets. Have Charley hook up the solar intake tubes to the spotter and replace the cathodes with pure thorium. I've got another idea."

"O.K., but it sounds foolish to me."

Channing laughed. "We'll stalemate him," he said bitterly, and explained to Walt. "I wonder when Murdoch will come this way?"

"It's but a matter of time," said Walt. "My bet is as soon as he can get here with that batch of fresh rats he's collected."

Walt's bet would have collected. Two days later, Hellion Murdoch flashed a signal into Venus Equilateral and asked for Channing.

"Hello, Hellion," answered Channing. "Haven't you learned to keep out of our way?"

"Not at all," answered Murdoch. "You won't try that betatron on me again. This ship is coated with four tenths of an inch of lithium metal, which according to the books will produce the maximum quantity of electrons under secondary emission. If not the absolute maximum, it is high enough to prevent your action."

"No," agreed Channing. "We won't try the betatron again. But, Murdoch, there are other things."

"Can they withstand these?" asked Murdoch. The turret swiveled and the triple-mount of tubes looked at Venus Equilateral.

"Might try," said Channing.

"Any particular place?" countered Murdoch.

"Hit the south end. We can best afford to lose that," answered Channing.

"You're either guessing, or hoping I won't fire, or perhaps praying that whatever you have for protection will work," said Murdoch flatly. "Otherwise you wouldn't talk so smooth."

"You black-hearted baby-killing rotter," snarled Don Channing. "I'm not chinning with you for the fun of it. You'll shoot anyway, and I want to see how good you are. Get it over with, Murdoch."

"What I have here is plenty good," said Murdoch. "Good enough. Do you know about it?"

"I can guess, but you tell me."

"Naturally," said Hellion. He explained in detail. "Can you best that?"

"We may not be able to outfire you," gritted Channing, "but we may be able to nullify your beam."

"Nonsense!" roared Murdoch, "Look, Channing, you'd best surrender."

"Never!"

"You'd rather die?"

"We'd rather fight it out. Come in and get us."

"Oh no. We'll just shoot your little Station full of holes. Like the average spaceship, your Station will be quite capable of handling communications even though the air is all gone. Filling us full of holes wouldn't do a thing; you see, we're wearing spacesuits."

"I guessed that. No, Murdoch, we have nothing to shoot at you this time. All we can do is to hold you off until you get hungry. You'll get hungry first, since we're self-sufficient."

"And in the meantime?"

"In the meantime we're going to try a few things out on your hull. I rather guess that you'll try out a few things on the Station, but at the present, you can't harm us and we can't harm you. Stalemate, Murdoch!"

"You're bluffing!" stormed Murdoch.

"Are you afraid to squirt that beam this way?" asked Channing tauntingly.
"Or do you know it will not work?"

"Why are you so anxious to get killed?"

"We're very practical, out here on Venus Equilateral," said Don. "There's no use in working further if you have something that is really good. We'd like to know our chances before we expend more effort along another line."

"That's not all—?"

"No. Frankly, I'm almost certain that your beam won't do a thing to Venus Equilateral."

"We'll see. Listen! Turretman! Are you ready?"

Faintly, the reply came, and Channing could hear it. "Ready!"

"Then fire all three. Pick your targets at will. One blast!"

The lights in Venus Equilateral brightened. The thousands of line-voltage meters went from one hundred and twenty-five to one hundred and forty volts, and the line-frequency struggled with the crystal-control and succeeded in making a ragged increase from sixty to sixty point one five cycles per second. The power-output meters on the transmitting equipment went up briefly, and in the few remaining battery-supply rooms, the overload and overcharge alarms clanged until the automatic adjusters justified the input against the constant load. One of the ten-kilowatt modulator tubes flashed over in the audio-room and was immediately cut from the operating circuit; the recording meters indicated that the tube had gone west forty-seven hours prior to its expiration date due to filament overload. A series of fluorescent lighting fixtures in a corridor of the Station that should have been dark because of the working hours of that section, flickered into life and woke several of the workers, and down in the laboratory, Wes Farrell swore because the fluctuating line had disrupted one of his experiments, giving him reason to doubt the result. He tore the thing down and began once more; seventy days work had been ruined.

"Well," said Channing cockily, "is that the best you can do?"

"You—!"

"You forgot," reminded Channing, "that we have been working with solar power, too. In fact, we discovered the means to get it. Go ahead and shoot at us, Murdoch. You're just giving us more power."

"Cease firing!" exploded Murdoch.

"Oh don't!" cheered Don. "You forgot that those tubes, if aligned properly, will actually cause bending of the energy-beam. We've got load-terminal tubes pointing at you, and your power-beam is bending to enter them. You did well, though. You were running the whole Station with plenty to spare. We had to squirt some excess into space. Your beams aren't worth the glass that's in them!"

"Stalemate, then," snarled Murdoch. "Now *you* come and get *us*. We'll leave. But we'll be back. Meanwhile, we can have our way with the shipping. Pilot! Course for Mars! Start when ready!"

The *Black Widow* turned and streaked from Venus Equilateral as Don Channing mopped his forehead. "Walt," he said, "that's once I was scared to death."

"Me, too. Well, we got a respite. Now what?"

"We start thinking."

"Right. But of what?"

"Ways and—Hello, Wes. What's the matter?"

Farrell entered and said: "They broke up my job. I had to set it up again, and I'm temporarily free. Anything I can do to help?"

"Can you dream up a space-gun?"

Farrell laughed. "That's problematical. Energy guns are something strange. Their output can be trapped and used to good advantage. What you need is some sort of projectile, I think."

"But what kind of projectile would do damage to a spaceship?"

"Obviously the normal kinds are useless. Fragmentation shells would pelt the exterior of the ship with metallic rain—if and providing you could get

them that close. Armor-piercing would work, possibly, but their damage would be negligible since hitting a spacecraft with a shell is impossible if the ship is moving at anything like the usual velocities. Detonation shells are a waste of energy, since there is no atmosphere to expand and contract. They'd blossom like roses and do as much damage as a tossed rose."

"No projectiles, then."

"If you could build a super-heavy fragmentation and detonation shell and combine it with armor-piercing qualities, and could hit the ship, you might be able to stop 'em. You'd have to pierce the ship, and have the thing explode with a terrific blast. It would crack the ship because of the atmosphere trapped in the hull—and should be fast enough to exceed the compressibility of air. Also it should happen so fast that the air leaving the hole made would not have a chance to decrease the pressure. The detonation would crack the ship, and the fragmentation would mess up the insides to boot, giving two possibilities. But if both failed and the ship became airless, they would fear no more detonation shells. Fragments would always be dangerous, however."

"So now we must devise some sort of shell—?"

"More than that. The meteor-circuits would intercept the incoming shell and it would never get there. What you'd need is a series of shells—say a hundred, all emitting the meteor-alarm primary signals, which would cause paralysis of the meteor-circuits. Then the big one, coming in at terrific velocity."

"And speaking of velocity," said Walt Franks. "The projectile and the rifle are out. We can get better velocity with a constant-acceleration drive. I say torpedoes!"

"Naturally. But the aiming? Remember, even though we crank up the drive to 50-G, it takes time to get to several thousand miles per second. The integration of a course would be hard enough, but add to it the desire of men to evade torpedoes—and the aiming job is impossible."

"We may be able to aim them with a device similar to the one Charley Thomas is working with. Murdoch said his hull was made of lithium?"

"Coated with," said Channing.

"Well. Set the alloy-selectivity disk to pure lithium, and use the output to steer the torpedo right down to the bitter end."

"Fine. Now the armor-piercing qualities."

"Can we drill?"

"Nope. At those velocities, impact would cause detonation, the combined velocities would look like a detonation wave to the explosive. After all, darned few explosives can stand shock waves that propagate through them at a few thousand miles per second."

"O.K. How do we drill?"

"We might drill electrically," suggested Farrell. "Put a beam in front?"

"Not a chance," grinned Channing. "The next time we meet up with Hellion Murdoch, he'll have absorbers ready for use. We taught him that one, and Murdoch is not slow to learn."

"So how do we drill?"

"Wes, is that non-arcing alloy of yours very conductive?"

"Slightly better than aluminum."

"Then I've got it! We mount two electrodes of the non-arcing alloy in front. Make 'em heavy and of monstrous current-carrying capacity. Then we connect them to a condenser made of Farrell's super-doooper dielectric."

"You bet," said Walt, grinning. "We put a ten microfarad condenser in front, only it'll be one hundred and thirty farads when we soak it in Farrell's super-dielectric. We charge it to ten thousand volts, and let it go."

"We've got a few experimental jobs," said Channing. "Those inerts. The drones we were using for experimental purposes. They were radio controlled, and can be easily converted to the aiming-circuits."

"Explosives?"

"We'll get the chemistry boys to brew a batch."

"Hm-m-m. Remind me to quit Saturday," said Walt. "I wonder how a ten farad condenser would drive one of those miniatures."

"Pretty well, I should imagine. Why?"

"Why not mount one of the miniatures on a gunstock and put a ten farad condenser in the handle? Make a nice side arm."

"Good for one shot, and not permanently charged. You'd have to cut your leakage down plenty."

"Could be. Well, we'll work on that one afterwards. Let's get that drone fixed."

"Let's fix up all the drones we have. And we'll have the boys wire up as many as they can of the little message-canisters. The whole works go at once at the same acceleration, with the little ones running interference for the big boy."

"Murdoch invited us to 'come and get him,'" said Channing in a hard voice. "That, I think we'll do!"

Four smoldering derelicts lay in absolute wreckage on or near the four great spaceports of the solar system. Shipping was at an unequalled standstill, and the communications beams were loaded with argument and recriminations and pleas as needed material did not arrive as per agreement. Three ships paid out one dollar each gross ton in order to take vital merchandise to needy parties, but the mine-run of shipping was unable to justify the terrific cost.

And then Don Channing had a long talk with Keg Johnson of Interplanetary Transport.

One day later, one of Interplanetary's larger ships took off from Canalopsis without having paid tribute to Murdoch. It went free—completely automatic—into the Martian sky and right into Murdoch's hands. The pirate gunned it into a molten mass and hurled his demands at the system once more, and left for Venus since another ship would be taking off from there.

In the *Relay Girl*, Don Channing smiled. "That finds Murdoch," he told Walt. "He's on the standard course for Venus from Mars."

"Bright thinking," commented Walt. "Bait him on Mars and then offer him a bite at Venus. When'll we catch him?"

"He's running, or will be, at about 3-G, I guess. We're roaring along at five and will pass Mars at better than four thousand miles per second. I think we'll catch and pass the *Black Widow* at the quarter-point, and Murdoch will be going at about nine hundred miles per. We'll zoom past, and set the finder on him, and then continue until we're safely away. If he gets tough, we'll absorb his output, though he's stepped it up to the point where a spacecraft can't take too much concentrated input."

"That's how he's been able to blast those who went out with absorbers?"

"Right. The stuff on the Station was adequate to protect, but an ordinary ship couldn't handle it unless the ship were designed to absorb and dissipate that energy. The beam-tubes would occupy the entire ship, leaving no place for cargo. Result: A toss-up between paying off and not carrying enough to make up the difference."

"This is Freddy," spoke the communicator. "The celestial globe has just come up with a target at eight hundred thousand miles."

"O.K., Freddy. That must be the *Black Widow*. How'll we pass her?"

"About thirty thousand miles."

"Then get the finders set on that lithium-coated hull as we pass."

"Hold it," said Walt. "Our velocity with respect to his is about three thousand. We can be certain of the ship by checking the finder-response on the lithium coating. If so, she's the *Black Widow*. Right from here, we can be assured. Jim! Check the finders in the torpedoes on that target!"

"Did," said Jim. "They're on and it is."

"Launch 'em all!" yelled Franks.

"Are you nuts?" asked Channing.

"Why give him a chance to guess what's happening? Launch 'em!"

"Freddy, drop two of the torpedoes and half of the interferers. Send 'em out at 10-G. We'll not put all our eggs in one basket," Channing said to Walt. "There might be a slip-up."

"It'll sort of spoil the effect," said Don, "But we're not here for effect."

"What effect?"

"That explosive will be as useless as a slab of soap," said Don. "Explosive depends for its action upon velocity—brother, there ain't no explosive built that will propagate at the velocity of our torpedo against Murdoch."

"I know," said Franks, smiling.

"Shall I yell 'Bombs away' in a dramatic voice?" asked Freddy Thomas.

"Are they?"

"Yup."

"Then yell," grinned Walt. "Look, Don, this should be pretty. Let's hike to the star-camera above and watch. We can use the double-telescope finder and take pix, too."

"It won't be long," said Channing grimly. "And we'll be safe since the interferers will keep Murdoch's gadget so busy he won't have time to worry us. Let's go."

The sky above became filled with a myriad of flashing spots as the rapidly-working meteor spotters coupled to the big turret and began to punch at the interferers.

The clangor of the alarm made Murdoch curse. He looked at the celestial globe and his heart knew real fear for the first time. This was no meteor shower, he knew from the random pattern. Something was after him, and Murdoch knew who and what it was. He cursed Channing and Venus Equilateral in a loud voice.

It did no good, that cursing. Above his head, the triply mounted turret danced back and forth, freeing a triple-needle of Sol's energy. At each pause another interferer went out in a blaze of fire and a shock-excitation of radio energy that blocked, temporarily, the finder circuits. And as the turret destroyed the little dancing motes, more came speeding into range to replace them, ten to one.

And then it happened. The finder-circuit fell into mechanical indecision as two interferers came at angles, each with the same intensity. The integrators ground together, and the forces they loosed struggled for control.

Beset by opposing impulses, the amplidyne in the turret stuttered, smoked, and then went out in a pungent stream of yellowish smoke that poured from its dust-cover in a high-velocity stream. The dancing of the turret stopped, and the flashing motes in the sky stopped with the turret's death.

One hundred and thirty farads, charged to ten thousand volts, touched the lithium-coated, aluminum side of Murdoch's *Black Widow*. Thirteen billion joules of electrical energy; thirty-six hundred kilowatt hours went against two inches of aluminum. At the three thousand miles per second relative velocity of the torpedo, contact was immediate and perfect. The aluminum hull vaporized under the million upon million of kilovolt-amperes of the discharge. The vaporized hull tried to explode, but was hit by the unthinkable velocity of the torpedo's warhead.

The torpedo itself crushed in front. It mushroomed under the millions of degrees Kelvin developed by the energy-release caused by the cessation of velocity. For the atmosphere within the *Black Widow* was as immobile and as hard as tungsten steel at its best.

The very molecules themselves could not move fast enough. They crushed together and in compressing brought incandescence.

The energy of the incoming torpedo raced through the *Black Widow* in a velocity wave that blasted the ship itself into incandescence. In a steep wave-front, the vaporized ship exploded in space like a supernova.

It blinded the eyes of those who watched. It overexposed the camera film and the expected pictures came out with one single frame a pure, seared black. The piffling, comparatively ladylike detonation of the System's best and most terrible explosive was completely covered in the blast.

Seconds later, the *Relay Girl* hurtled through the sky three thousand miles to one side of the blast. The driven gases caught the *Girl* and stove in the upper observation dome like an eggshell. The *Relay Girl* strained at her girders, and sprung leaks all through the rigid ship, and after rescuing Don Channing and Walt Franks from the wreckage of the observation dome, the men spent their time welding cracks until the *Relay Girl* landed.

It was Walt who put his finger on the trouble. "That was period for Murdoch," he said. "But Don, the stooge still runs loose. We're going to be forced to take over Mark Kingman before we're a foot taller. He includes

Terran Electric, you know. That's where Murdoch got his machine work done."

"Without Murdoch, Kingman is fairly harmless," said Don, objecting. "We'll have no more trouble from him."

"You're a sucker, Don. Kingman will still be after your scalp. You mark my words."

"Well, what are you going to do about it?"

"Nothing for the present. I've got some unfinished business to attend to at Lincoln Head. Mind?"

THE END.

*** END OF THE PROJECT GUTENBERG EBOOK THE FIRING LINE

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